

## Naive Bayes Classifier

### Conditional Probability

$\Rightarrow A, B$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

where,  $P(B) \neq 0$

e.g.,

Two dice are rolled together.

$\{(1,1), (1,2), \dots, (2,1), (2,2), \dots, (6,6)\}$

total be 36.

$D_1$

|         | 1 | 2 | 3 | 4  | 5  | 6  |
|---------|---|---|---|----|----|----|
| 1       | 2 | 3 | 4 | 5  | 6  | 7  |
| 2       | 3 | 4 | 5 | 6  | 7  | 8  |
| $D_2$ 3 | 4 | 5 | 6 | 7  | 8  | 9  |
| 4       | 5 | 6 | 7 | 8  | 9  | 10 |
| 5       | 6 | 7 | 8 | 9  | 10 | 11 |
| 6       | 7 | 8 | 9 | 10 | 11 | 12 |

$\therefore D_1 = 5$  given  $D_1 + D_2 \leq 10$

$\uparrow$   
A

$\uparrow$   
B

$$\therefore P(D_1 = 5 | D_1 + D_2 \leq 10) = \frac{5}{33} = P(A|B)$$

OR,

$$P(A|B) = \frac{\frac{5}{36}}{\frac{33}{36}} = \frac{5}{33}$$

### Independent Events in Prob.

$\Rightarrow$  if A and B are two events such that

$$P(A \cap B) = P(A) * P(B)$$

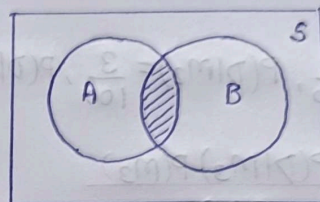
then it be independent events,

e.g.,

(i) Tossing two coins together

(ii) Rolling two dice together.

$\Rightarrow A/B$ ,



$$P(A|B) = \frac{\pi(A \cap B)}{\pi(B)}$$

and,

$$P(A) = \frac{P(A \cap B)}{P(B)} = \frac{\pi(A \cap B)}{\pi(S)} \times \frac{\pi(S)}{\pi(B)}$$

$$= \frac{\pi(A \cap B)}{\pi(B)}$$

$\therefore P(A|B) = P(A) \rightarrow$  if they are ind. events

$\Rightarrow$  Mutually exclusive events:

$$P(A \cap B) = 0$$

and

$$P(A|B) = 0$$

e.g., if we take right given we have already taken left.

### Bayes Theorem

$\Rightarrow$  if A and B are two events then,

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

given,  $P(B) \neq 0$

e.g., in a Factory there are three machines ( $M_1, M_2, M_3$ ) where  $M_1$  is producing 20%,  $M_2$  is producing 30% and  $M_3$  is producing 50%. While defective production % of them be 5%, 3% and 1% respectively.

(i) Sample taken is defective and is from  $M_3$  what be its probability?

$$P(M_1) = \frac{20}{100} = \frac{1}{5}, P(M_2) = \frac{30}{100} = \frac{3}{10}, P(M_3) = \frac{1}{2}$$

and,

$$P(D|M_1) = \frac{5}{100} = \frac{1}{20}, P(D|M_2) = \frac{3}{100}, P(D|M_3) = \frac{1}{100}$$

$$\therefore P(M_3|D) = \frac{P(D|M_3)P(M_3)}{P(D)}$$

$$\begin{aligned} \therefore P(D) &= P(D \cap M_1) + P(D \cap M_2) + P(D \cap M_3) \\ &= P(D|M_1)P(M_1) + P(D|M_2)P(M_2) \\ &\quad + P(D|M_3)P(M_3) \end{aligned}$$

### Maths Behind Naive Bayes

$\Rightarrow$  Let,  $X = \{x_1, x_2, \dots, x_n\}$   
 $C_k = \{C_1, C_2, \dots, C_K\} \rightarrow \text{target}$

$$P(C_k|X) = \frac{P(X|C_k)P(C_k)}{P(X)}$$

denominator is removed as it has no contribution in results

$$\therefore P(A|B) = \frac{P(A \cap B)}{P(B)} \Rightarrow P(A \cap B) = P(A|B)P(B)$$

$$\begin{aligned} \therefore P(C_k|X) &= P(X \cap C_k) \text{ or } P(X, C_k) \\ &= P(\overset{A}{x_1}, \overset{B}{x_2, x_3, \dots, x_n}, C_k) \\ &= P(x_1 | x_2, x_3, x_4, \dots, x_n, C_k) P(x_2, x_3, x_4, \dots, x_n, C_k) \\ &= D.X.P(x_2, x_3, \dots, x_n, C_k) \\ &= G.B.P(x_1, \dots, x_n, C_k) \\ &= P(x_1 | x_2, x_3, \dots, x_n, C_k) P(x_2 | x_3, x_4, \dots, x_n, C_k) \\ &\quad P(x_3 | x_4, x_5, \dots, x_n, C_k) \dots \\ &\quad P(x_{n-1} | x_n, C_k) P(x_n | C_k) P(C_k) \end{aligned}$$

By Naive assumption,

$$\begin{aligned} P(C_k|X) &= P(x_1|C_k)P(x_2|C_k)P(x_3|C_k) \dots \\ &\quad P(x_n|C_k)P(C_k) \\ &= P(C_k) \prod_{i=1}^n P(x_i|C_k) \end{aligned}$$

Now, by Maximum a posteriori Rule (MAP):

$$\hat{y} = \underset{K \in \{1, 2, \dots, K\}}{\text{argmax}} P(C_k) \prod_{i=1}^n P(x_i|C_k)$$

|    |    |    |   |   |   |
|----|----|----|---|---|---|
| 2  | 2  | 1  | 3 | 5 | 1 |
| F  | 2  | 2  | 1 | 3 | 5 |
| 8  | F  | 2  | 2 | 1 | 3 |
| 2  | 8  | F  | 2 | 2 | 1 |
| 10 | 2  | 8  | F | 2 | 1 |
| 11 | 10 | 2  | 8 | F | 2 |
| 15 | 11 | 10 | 2 | 8 | F |